

Noah MARSHALL

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in linkedin | github | webpage

EDUCATION

McGill University

Sept. 2023 - present

PhD Candidate in Mathematics and Statistics

- Supervisor: Dr. Elliot Paquette
- Research Topics: random matrix theory, stochastic processes, high-dimensional probability and optimization

McGill University

2020-2023

Master of Science in Mathematics and Statistics

- Supervisor: Dr. Adam Oberman
- Cumulative GPA: 4.0/4.0
- Associated with Mila – Quebec Artificial Intelligence Institute

The University of British Columbia

2015-2019

Bachelor of Science - Honours in Mathematics and Minor in Data Science

- Cumulative GPA: 4.33/4.33

SKILLS

MATHEMATICS	Probability stochastic processes random matrix theory high-dimensional statistics optimization
MODELLING	Covariance estimation in high dimensions particle filters & data assimilation optimizer & training dynamics scaling laws

RESEARCH EXPERIENCE

Microsoft + Nuance

2023 Summer

NLP Research Intern

- Implemented generative AI tools for customer service
- Augmented large language models using a knowledge base retriever

Phyla

2021 Summer

Machine Learning Research Intern

- Developed a graph neural network for classification of irritable bowel diseases (IBD)
- Implemented a pipeline to run, test, and log experimental results
- Developed a custom cross validation method to reduce the impact of batch effects

The University of British Columbia

2020-2021

Research Assistant 2

- Analyzed and developed a differential equation model coupling the dynamics of opinion and disease
- Find that more reactive populations can experience smaller peak epidemic size, but more epidemic waves and more cases

Mathematics and Climate Research Network

2019-2021

Student Researcher

- Collaborated with an international group of students and professors to develop a novel data assimilation method
- Investigated the use of model and data dimension reduction methods to improve the effectiveness of particle filters

PUBLICATIONS

- [arXiv 2026] **Phases of Muon: When Muon Eclipses SignSGD** — E. Paquette, **Marshall**, Benigni, Wang, Agarwala, C. Paquette
- [ICML 2025] **Exact Risk Curves of signSGD in High-Dimensions: Quantifying Preconditioning and Noise-Compression Effects** — Xiao, **Marshall**, Agarwala, Paquette
- [ICLR 2025] **To Clip or Not to Clip: the Dynamics of SGD with Gradient Clipping in High-Dimensions** — **Marshall**, Xiao, Agarwala, Paquette
- [ICLR 2022] **FairCal: Fairness Calibration for Face Verification** — Salvador, Cairns, Voleti, **Marshall**, Oberman
- [AIMS Math. 2022] **Transient prophylaxis and multiple epidemic waves** — Tyson, **Marshall**, Baumgaertner
- [Comput. Math. Appl. 2022] **Model and Data Reduction for Data Assimilation: Particle Filters Employing Projected Forecasts and Data with Application to a Shallow Water Model** — Albarakati, Budišić, Crocker, Glass-Klaiber, Iams, Maclean, **Marshall**, Roberts, Van Vleck (equal contribution, sorted by last name)

TEACHING AND MENTORSHIP

Course Instructor — Intermediate Calculus

2026

McGill University

Directed Reading Program — mentor

2021-present

McGill University. Gaussian processes on manifolds, generative diffusion models, probabilistic graphical models, tree indexed stochastic processes, percolations on neural networks

Graduate Teaching Assistant

2022-present

McGill University. Introduction to stochastic processes, Calculus 1

AWARDS

Canadian Graduate Scholarship (NSERC)

present

Institut des sciences mathématiques Graduate Scholarship

present

NSERC Undergraduate Student Research Award

2018